

# E-Commerce User Behavior Analysis

Conversion Funnel & Marketing Channel Performance Report

100,000

TOTAL VISITORS

12,250

COMPLETED ORDERS

12.25%

SITE CONV. RATE

## Conversion Funnel Breakdown

Analysis of the primary customer journey: View → Cart → Purchase.

| Funnel Stage | Total Users | Step Conversion | Drop-off Rate | Cumulative Conv. |
|--------------|-------------|-----------------|---------------|------------------|
| 1. View      | 100,000     | 100.0%          | 0.0%          | 100.0%           |
| 2. Cart      | 35,000      | 35.0%           | 65.0%         | 35.0%            |
| 3. Purchase  | 12,250      | 35.0%           | 65.0%         | 12.3%            |

**The "Leaky Bucket" Phenomenon:** The primary friction point occurs between View → Cart, where 65% of traffic is lost. This indicates potential issues with landing page relevance or product presentation.

## Marketing Acquisition Performance

| Channel        | Views  | Carts  | Purchases | V2C % | C2P % | Overall % |
|----------------|--------|--------|-----------|-------|-------|-----------|
| Paid Ads       | 40,000 | 12,000 | 3,000     | 30.0% | 25.0% | 7.5%      |
| Organic Search | 30,000 | 12,000 | 5,400     | 40.0% | 45.0% | 18.0%     |

| Channel      | Views  | Carts | Purchases | V2C % | C2P % | Overall % |
|--------------|--------|-------|-----------|-------|-------|-----------|
| Social Media | 20,000 | 8,000 | 2,400     | 40.0% | 30.0% | 12.0%     |
| Email/Direct | 10,000 | 3,000 | 1,450     | 30.0% | 48.3% | 14.5%     |

**Key Discovery:** Organic Search shows the highest intent with an 18% conversion rate, while Paid Ads—despite high volume—convert at only 7.5%, suggesting a need for better keyword targeting.

## Strategic Recommendations

- **Optimize Paid Ad Targeting:** Refine audiences to focus on high-intent long-tail keywords.
- **Implement Cart Recovery:** Use automated email sequences for the 65% of users who abandon carts.
- **Streamline Mobile Checkout:** Reduce friction for Social Media traffic by simplifying mobile forms.
- **Scale SEO Operations:** Invest more in content that drives high-converting organic traffic.

## Analytical Implementation (Python)

```
import pandas as pd

# Data structure for replication
data = {
    'Channel': ['Paid Ads', 'Organic Search', 'Social Media', 'Email/Direct'],
    'Views': [40000, 30000, 20000, 10000],
    'Carts': [12000, 12000, 8000, 3000],
    'Purchases': [3000, 5400, 2400, 1450]
}

df = pd.DataFrame(data)

# Compute KPIs
df['V2C_Rate'] = (df['Carts'] / df['Views']) * 100
df['C2P_Rate'] = (df['Purchases'] / df['Carts']) * 100
df['Overall_Conv'] = (df['Purchases'] / df['Views']) * 100
```

```
print(df.round(2))
```

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Simulated E-Commerce Dataset Analysis